

**THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

MAT 3C 03—MATHEMATICS

(2014—2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Part A (Objective Type)

*Answer all the **twelve** questions.
Each question carries 1 mark.*

1. Define ordinary differential equation.
2. The solution of the differential equation $y' - 1 + y^2$ is _____.
3. The degree of the differential equation $\frac{dy}{dx} = -2 \sin x \cos x$ is _____.
4. The rank of the matrix $A = \begin{bmatrix} 0 & 2 \\ 0 & 5 \end{bmatrix}$ is _____.
5. State True or False : The following two matrices are equivalent :

$$\begin{bmatrix} 5 & 5 & -5 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 3 & -3 & 3 & -7 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 3 & -3 & 3 & -7 \end{bmatrix}.$$

6. The characteristic matrix of the matrix $\begin{bmatrix} 0 & 1 \\ 4 & 0 \end{bmatrix}$ is _____.
7. Find the resultant of the vectors $\mathbf{p} = [4, -2, -3]$, $\mathbf{q} = [8, 8, 1]$, and $\mathbf{u} = [-12, -6, 2]$.
8. For the vectors $\mathbf{a} = [1, 3, 2]$, $\mathbf{b} = [2, 0, -5]$, and $\mathbf{c} = [4, -2, 1]$, find $(\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$.

Turn over

9. The vector function $\mathbf{r}(t) = (4 + t)\mathbf{i} + (2 + t)\mathbf{j}$ represents _____.
10. Let $\vec{r}(t) = 5t^2 \vec{k}$ be the position vector of a moving particle, where $t \geq 0$ is time. Then the acceleration vector of the moving particle is _____.
11. If $\vec{v} = e^x (\cos y \mathbf{i} + \sin y \mathbf{j})$ then $\text{div } \vec{v} = \text{_____}$.
12. When we say that a vector valued function is conservative ?

(12 × 1 = 12 marks)

Part B (Short Answer Type)

*Answer any nine questions.
Each question carries 2 marks.*

13. Verify that $y = e^{4x}$ is a solution of the differential equation $\frac{dy}{dx} = 4y$.
14. Solve the initial value problem $2xy' - 3y = 0$; $y(1) = 4$.
15. Show that the equation :

$$ydx + xdy = 0$$

is exact and solve it.

16. Reduce the matrix $\begin{bmatrix} 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ to its normal form.

17. Solve completely the system of equations :

$$x + 3y - 2z = 0$$

$$2x - y + 4z = 0$$

$$x - 11y + 14z = 0.$$

18. Find the eigen values of :

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{bmatrix}$$

19. Prove that $(\mathbf{u} + \mathbf{v})' = \mathbf{u}' + \mathbf{v}'$

20. Find $\frac{df}{ds}$ in the direction of the vector $4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ at the point $(1, 1, 2)$ if $f(x, y, z) = x^2 + y^2 - z$.

21. Show that $\mathbf{F}(x, y) = (\cos y + y \cos x) \mathbf{i} + (\sin x - x \sin y) \mathbf{j}$ is a conservative vector field.
22. Find the unit tangent vector at a point t to the curve $\mathbf{r} = a \cos t \mathbf{i} + a \sin t \mathbf{j}$.
23. Find unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$.
24. Verify that $w = x^2 - y^2$ satisfies Laplace's equation $\nabla^2 w = 0$.

(9 × 2 = 18 marks)

Part C (Short Essays)

Answer any **six** questions.
Each question carries 5 marks.

25. Write in the linear form and then solve $\sin 2x \frac{dy}{dx} = y + \tan x$.
26. Determine the rank of the following matrix, by reducing to echelon form :

$$\begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 1 & a & b & 0 \\ 0 & c & d & 1 \end{bmatrix}.$$

27. Show that the system of equations :

$$\begin{aligned} x + 2y + z &= 2 \\ 3x + y - 2z &= 1 \\ 4x - 3y - z &= 3 \\ 2x + 4y + 2z &= 4 \end{aligned}$$

is consistent and hence solve them.

28. Find the eigen values and the eigen vector corresponding to the largest eigen value of the matrix

$$\begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}.$$

29. Using Cayley-Hamilton theorem find the inverse of $A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$.

30. Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ with

$$\mathbf{F}(r) = 5z\mathbf{i} + xy\mathbf{j} + x^2z\mathbf{k}$$

along the straight - line segment $C : t\mathbf{i} + t\mathbf{j} + t\mathbf{k}, 0 \leq t \leq 1$.

31. Evaluate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n} \, dA$ by the divergence theorem where $\mathbf{F} = [x^2, 0, z^2]$ and S is

the surface of the box given by the inequalities $|x| \leq 1, |y| \leq 3, |z| \leq 2$.

32. Evaluate the flux integral $\iint_S \mathbf{F} \cdot \mathbf{n} \, dA$ where $\mathbf{F} = [e^{2y}, e^{-2z}, e^{2x}]$, and

$$S : \mathbf{r} = [3 \cos u, 3 \sin u, v], 0 \leq u \leq \frac{1}{2}\pi, 0 \leq v \leq 2.$$

33. Apply Green's theorem to evaluate $\oint_C (2x^2 - y^2) dx + (x^2 + y^2) dy$, where C is the boundary of the area enclosed by the x -axis and the upper half of the circle $x^2 + y^2 = a^2$.

(6 × 5 = 30 marks)

Part D

*Answer any two questions.
Each question carries 10 marks.*

34. Find the orthogonal trajectories of the family of curves $x^2 + y^2 = c^2$.
35. Investigate for what values of a, b the system of equations :

$$x + y + 2z = 2$$

$$2x - y + 3z = 10$$

$$5x - y + az = b$$

have unique solution.

36. Calculate the line integral $\oint_C \mathbf{F} \cdot \mathbf{r}'(s) \, ds$ using Stoke's theorem, where $\mathbf{F} = [-5y, 4x, z]$, and C is the circle $x^2 + y^2 = 4, z = 1$.

(2 × 10 = 20 marks)

**THIRD SEMESTER (CUCBCSS—UG) DEGREE [SPECIAL] EXAMINATION
NOVEMBER 2019**

Mathematics

MAT 3C 03—MATHEMATICS

Time : Three Hours

Maximum : 80 Marks

Part A (Objective Type)

*Answer all the twelve questions.
Each question carries 1 mark.*

1. Define differential equation.
2. The solution of the differential equation $y' = ky$ is _____.
3. The order of the differential equation $\frac{dy}{dx} = \sin x$ is _____.
4. The rank of the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix}$ is _____.
5. State True or False : The following two matrices are equivalent :

$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 3 & -3 & 3 & -7 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 3 & -3 & 3 & -7 \end{bmatrix}.$$

6. The characteristic matrix of the matrix $\begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$ is _____.
7. Let $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = 3\mathbf{j}$, and $\mathbf{c} = 4\mathbf{i} + \mathbf{j} - \mathbf{k}$. Then $(\mathbf{a} + \mathbf{b}) + \mathbf{c} =$ _____.
8. For the vectors $\mathbf{a} = [1, 3, 2]$, $\mathbf{b} = [2, 0, -5]$ and $\mathbf{c} = [4, -2, 1]$, find $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c})$.

9. The vector function $\mathbf{r}(t) = 3 \cos t \mathbf{i} - 4 \sin t \mathbf{j}$ ($0 \leq t \leq 2\pi$) represents _____.
10. Let $\mathbf{r}(t) = 4t^2 \mathbf{k}$ be the position vector of a moving particle, where $t \geq 0$ is time. The acceleration vector of the moving particle is _____.
11. If $\mathbf{v} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ then $\text{div } \mathbf{v} = \text{_____}$.
12. When we say that a vector valued function is irrotational ?

(12 × 1 = 12 marks)

Part B (Short Answer Type)

*Answer any nine questions.
Each question carries 2 marks.*

13. Verify that $y = e^{3x}$ is a solution of the differential equation $\frac{dy}{dx} = 3y$.

14. Solve the initial value problem $\frac{dy}{dx} - y \tan 2x = 0$; $y(0) = 2$.

15. Show that the equation :

$$(1 + 4xy + 2y^2) dx + (1 + 4xy + 2x^2) dy = 0$$

is exact and solve it.

16. Reduce the matrix $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{bmatrix}$ to its normal form.

17. Solve completely the system of equations :

$$x + 3y - 2z = 0$$

$$2x - y + 4z = 0$$

$$x + 11y + 14z = 0$$

18. Find the eigen values of :

$$\begin{bmatrix} 5 & 1 & 1 \\ 1 & 3 & 1 \end{bmatrix}$$

19. Prove that $(\mathbf{u} \times \mathbf{v})' = \mathbf{u}' \times \mathbf{v} + \mathbf{u} \times \mathbf{v}'$.
20. What is the maximum possible $\frac{df}{ds}$ at the point $(1, 4, 2)$, if $f(x, y, z) = x^2 + y^2 - z$.
21. Show that $\mathbf{F} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$ is a conservative vector field.
22. Find the unit tangent vector to the curve $x = t^2 + 1, y = 4t - 3, z = 2t^2 - 6t$ at the point $t = 2$.
23. Find unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$.
24. Verify that $w = x^2 - y^2$ satisfies Laplace's equation $\nabla^2 w = 0$.
- (9 × 2 = 18 marks)

Part C (Short Essays)

*Answer any six questions.
Each question carries 5 marks.*

25. Solve the linear differential equation $y' - y = e^{2x}$.
26. Determine the rank of the following matrix, by reducing to echelon form

$$\begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 1 & a & b & 0 \\ 0 & c & d & 1 \end{bmatrix}$$

27. Test the following system of equations for consistency and solve it, if it is consistent.

$$x + y + z = 6$$

$$x + y + 2z = 5$$

$$3x + y + z = 8$$

28. Find the eigen values and the eigen vector corresponding to the largest eigen value of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & -1 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

29. Using Cayley-Hamilton theorem, find the inverse of:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

30. If $\mathbf{F} = (3x^2 + 6y)\mathbf{i} - 14yz\mathbf{j} + 20xz^2\mathbf{k}$, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is the curve from $(0, 0, 0)$ to $(1, 1, 1)$ with parametric form $x = t, y = t^2, z = t^3$.
31. Evaluate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n} dA$ by the divergence theorem where $\mathbf{F} = [x^2, 0, z^2]$ and S is the surface of the box given by the inequalities $|x| \leq 1, |y| \leq 3, |z| \leq 2$.
32. Evaluate the flux integral $\iint_S \mathbf{F} \cdot \mathbf{n} dA$ where $\mathbf{F} = [3x^2, y^2, 0]$, and $S: \mathbf{r} = [u, v, 2u + 3v], 0 \leq u \leq 2, -1 \leq v \leq 1$.
33. Verify Green's theorem in the plane for $\oint_C (xydx + x^2dy)$, where C is the curve enclosing the region bounded by the parabola $y = x^2$ and the line $y = x$.

(6 × 5 = 30 marks)

Part D

*Answer any two questions.
Each question carries 10 marks.*

34. Solve $\frac{dy}{dx} + x \sin^2 y = x^3 \cos^2 y$.
35. Investigate for what values of a, b the system of equations :
- $$\begin{aligned} x + y + 2z &= 2 \\ 2x - y + 3z &= 10 \\ 5x - y + az &= b \end{aligned}$$
- have no solution.
36. Verify $\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dA = \oint_C \mathbf{F} \cdot \mathbf{r}'(s) ds$ where $\mathbf{F} = [z^2, 5x, 0]$, S is the square $0 \leq x \leq 1, 0 \leq y \leq 1, z = 1$ and C is the boundary of S .

(2 × 10 = 20 marks)